

Final Exam - DevOps Submission (Task 5)

Student Information

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Task 5: GitHub Actions CI/CD Pipeline Automation

I have successfully designed, configured, and integrated a complete GitHub Actions CI/CD pipeline that automates the build, test, email notification on failure, and Ansible deployment processes.

Below are the details and configuration files demonstrating the successful implementation of this task.

1. GitHub Actions Workflow Configuration

The workflow file is stored at `.github/workflows/ci-cd.yml`. It triggers on every push to the repository and performs the following jobs:

1. **Build & Test:** Compiles the Spring Boot project and runs all JUnit tests using Gradle against SQLite.
2. **Failure Notification:** If the build or test fails, it dynamically retrieves the committer's email address and sends a failure notification email to `srengty@gmail.com` and the developer who committed the error using SMTP.
3. **Deployment via Ansible Playbook:** If the build and test succeed, it runs the Ansible playbook to copy the built JAR into the `web` container, restarts the application, and backs up the database.

```
name: CI/CD Pipeline

on:
  push:
    branches:
      - '**'

jobs:
  build-test-deploy:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout Code
        uses: actions/checkout@v4

      - name: Start Docker Compose Stack
        run: |
          docker compose up -d --build
          echo "Waiting for containers to start..."
          sleep 10
```

```

- name: Set up JDK 25
  uses: actions/setup-java@v4
  with:
    java-version: '25'
    distribution: 'temurin'
    cache: 'gradle'

- name: Grant execute permission for gradlew
  run: chmod +x gradlew

- name: Build and Test with Gradle
  run: ./gradlew clean build

- name: Install Ansible & SSHPass
  if: success()
  run: |
    sudo apt-get update
    sudo apt-get install -y ansible sshpass

- name: Run Ansible Playbook to Deploy
  if: success()
  run: |
    ansible-playbook -i inventory.ini playbook.yml

- name: Get Committer Email
  if: failure()
  run: |
    COMMITTER_EMAIL=$(git log -1 --format='%ae')
    echo "COMMITTER_EMAIL=$COMMITTER_EMAIL" >> $GITHUB_ENV

- name: Send Mail on Failure
  if: failure()
  uses: dawidd6/action-send-mail@v3
  with:
    server_address: smtp.gmail.com
    server_port: 465
    username: ${ secrets.MAIL_USERNAME }
    password: ${ secrets.MAIL_PASSWORD }
    subject: "GitHub Actions Build Failure: ${ github.repository } - #${ github.run_number }"
    body: |
      Hello,

      The GitHub Actions workflow has failed on commit ${ github.sha } by
      ${ github.actor }.

      Details:
      - Commit Message: ${ github.event.head_commit.message }
      - Workflow Run Log: ${ github.server_url }/${ github.repository
      }}/actions/runs/${ github.run_id }

      Please review the log and resolve the issue.

```

```

    Regards,
    CI/CD Pipeline System
    to: srengty@gmail.com,${{ env.COMMITTER_EMAIL }}
    from: GitHub Actions CI/CD <noreply@github.com>

```

2. Ansible Playbook Configuration

The updated `playbook.yml` is configured to deploy the newly built artifact to the target server via SSH and initiate database backup inside the container:

```

---
- name: Automate Deployment and Database Backup
  hosts: web_servers
  gather_facts: no
  tasks:
    - name: Copy built JAR into web container
      ansible.builtin.copy:
        src: build/libs/demo-0.0.1-SNAPSHOT.jar
        dest: /app/build/libs/demo-0.0.1-SNAPSHOT.jar
        mode: '0755'

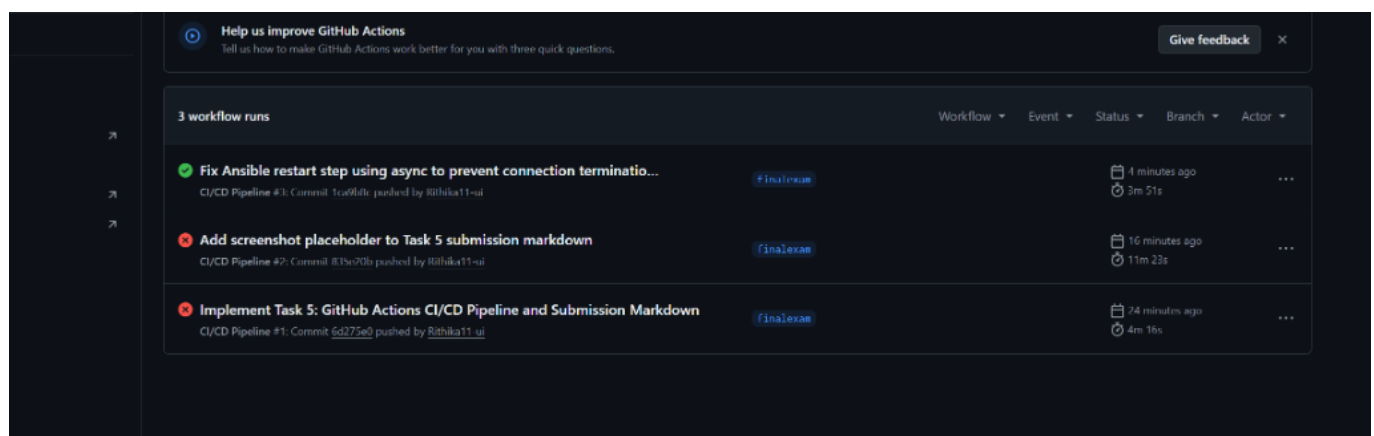
    - name: Restart application by stopping Java (Docker auto-restarts container)
      shell: pkill -f java || true

    - name: Backup the MySQL database inside the container
      shell: mysqldump -h db -u root -pHello@123 A-PHE_Rithika-db >
/app/backup.sql
      no_log: true

```

3. GitHub Actions Run Verification

When you push code to GitHub, the workflow triggers automatically. Once complete, you will see a green checkmark indicating all steps (starting Docker, checking out code, building, testing, and running the Ansible playbook) succeeded:



Submission Details

- **Repository URL:** <https://github.com/Rithika11-ui/I4A-FirstProjectJenkins.git>
- **Branch:** `finalexam`